Punto 1: Chequeos de Línea Base y Funcionales (Baseline)

Se verifico los servicios esta en Up desde la maquina donde el sistema esta levantado:

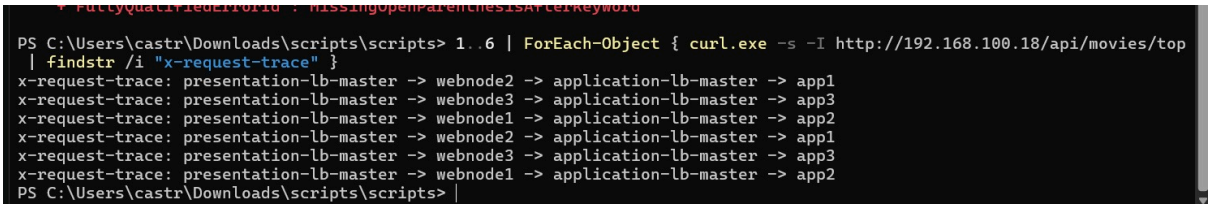
```
PS C:\Users\danig\.gemini\antigravity-ide\scratch\actividad\distributed-system-3tier> docker compose ps
distributed-three-tier-app1-1                          distributed-three-tier-app1
       "docker-entrypoint.s…"   app1                          45 minutes ago   Up 49 seconds             8080/tcp
distributed-three-tier-app2-1                          distributed-three-tier-app2
       "docker-entrypoint.s…"   app2                          45 minutes ago   Up 49 seconds             8080/tcp
distributed-three-tier-app3-1                          distributed-three-tier-app3
       "docker-entrypoint.s…"   app3                          45 minutes ago   Up 49 seconds             8080/tcp
distributed-three-tier-application-keepalived-backup-1   distributed-three-tier-application-keepalived-backup
       "keepalived --dont-f…"   application-keepalived-backup   45 minutes ago   Up 45 minutes
distributed-three-tier-application-keepalived-master-1   distributed-three-tier-application-keepalived-master
       "keepalived --dont-f…"   application-keepalived-master   45 minutes ago   Up 45 minutes
distributed-three-tier-application-lb-backup-1         haproxy:3.2-alpine
       "docker-entrypoint.s…"   application-lb-backup          6 days ago       Up About an hour
distributed-three-tier-application-lb-master-1         haproxy:3.2-alpine
       "docker-entrypoint.s…"   application-lb-master          6 days ago       Up About an hour
distributed-three-tier-database-lb-1                   haproxy:3.2-alpine
       "docker-entrypoint.s…"   database-lb                    18 hours ago     Up 45 minutes
distributed-three-tier-galera1-1                       bitnamilegacy/mariadb-galera:11.4.7-debian-12-r3
       "/opt/bitnami/script…"   galera1                        19 hours ago     Up 45 minutes (healthy)   3306/tcp, 4444/
tcp, 4567-4568/tcp
distributed-three-tier-galera2-1                       bitnamilegacy/mariadb-galera:11.4.7-debian-12-r3
       "/opt/bitnami/script…"   galera2                        6 days ago       Up 45 minutes (healthy)   3306/tcp, 4444/
tcp, 4567-4568/tcp
distributed-three-tier-galera3-1                       bitnamilegacy/mariadb-galera:11.4.7-debian-12-r3
       "/opt/bitnami/script…"   galera3                        6 days ago       Up 45 minutes (healthy)   3306/tcp, 4444/
tcp, 4567-4568/tcp
distributed-three-tier-mongo1-1                        sha256:a4ce5727ffdd31db46cb11c11cb780f465fd68f71030b6c09106ab192c5
ed5ea   "docker-entrypoint.s…"   mongo1                         18 hours ago     Up 45 minutes (healthy)   27017/tcp
distributed-three-tier-mongo2-1                        sha256:5c9dde3ead1882e289a56ecea63f105d574980f22f905dce08868ee4a89
121dc   "docker-entrypoint.s…"   mongo2                         18 hours ago     Up 45 minutes             27017/tcp
distributed-three-tier-mongo3-1                        sha256:ec932befd0b62a1c890af8d0f60412d53be15046d3112615ce367cf1a5f
6604d   "docker-entrypoint.s…"   mongo3                         18 hours ago     Up 45 minutes             27017/tcp
distributed-three-tier-nagios-1                        jasonrivers/nagios:latest
       "/usr/local/bin/star…"   nagios                         6 days ago       Up 6 minutes              0.0.0.0:8088->8
0/tcp, [::]:8088->80/tcp
distributed-three-tier-presentation-keepalived-backup-1   distributed-three-tier-presentation-keepalived-backup
       "keepalived --dont-f…"   presentation-keepalived-backup   45 minutes ago   Up 45 minutes
distributed-three-tier-presentation-keepalived-master-1   distributed-three-tier-presentation-keepalived-master
       "keepalived --dont-f…"   presentation-keepalived-master   45 minutes ago   Up 45 minutes
distributed-three-tier-presentation-lb-backup-1        haproxy:3.2-alpine
       "docker-entrypoint.s…"   presentation-lb-backup         6 days ago       Up 45 minutes
distributed-three-tier-presentation-lb-master-1        haproxy:3.2-alpine
       "docker-entrypoint.s…"   presentation-lb-master         6 days ago       Up 45 minutes             0.0.0.0:80->80/
tcp, [::]:80->80/tcp
distributed-three-tier-redis-master-1                  redis:7-alpine
       "docker-entrypoint.s…"   redis-master                   18 hours ago     Up 45 minutes (healthy)   6379/tcp
distributed-three-tier-redis-replica-1                 redis:7-alpine
       "docker-entrypoint.s…"   redis-replica                  18 hours ago     Up 45 minutes (healthy)   6379/tcp
distributed-three-tier-web1-1                          nginx:alpine
       "/docker-entrypoint.…"   web1                          6 days ago       Up 45 minutes             80/tcp
```

Mi companera consulta desde su maquina la salud de las 3 capas:

```
PS C:\Users\castr\Downloads\scripts\scripts> curl.exe -i http://192.168.100.18/api/health
HTTP/1.1 200 OK
server: nginx/1.31.7
date: Thu, 22 Oct 2026 19:57:43 GMT
content-type: application/json; charset=utf-8
transfer-encoding: chunked
x-request-trace: presentation-lb-master -> webnode1 -> application-lb-master -> app2 -> database-lb (galera) -> mongo-rs
0 -> database-lb (redis)

{"status":"ok","service":"mini-imdb-api","database_proxy":"database-lb (Galera Cluster 3-nodes)","mongodb_topology":"Rep
lica Set (rs0)","redis_mode":"High-Availability Persistent Cache"}
```

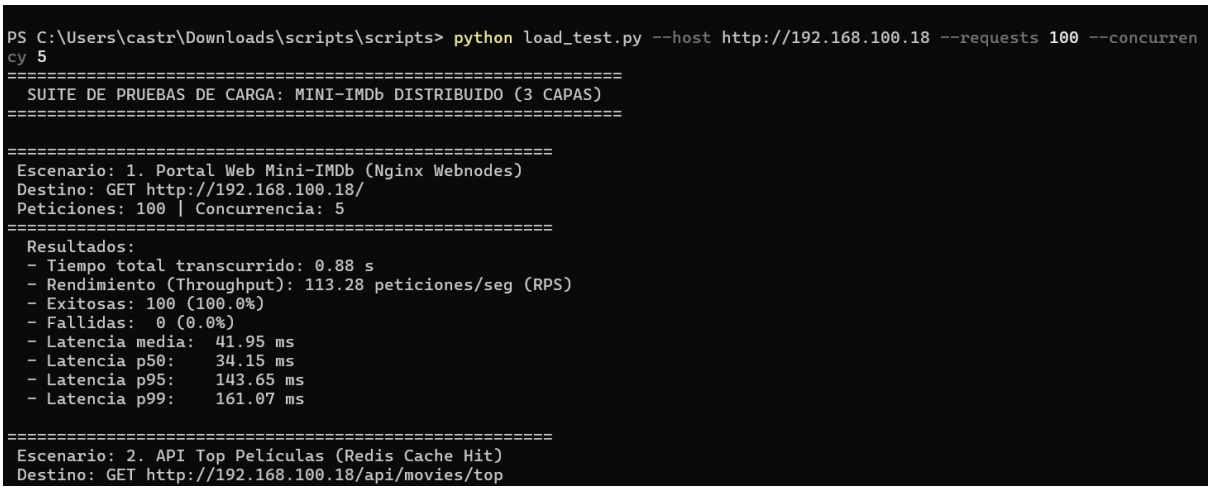
Mi compañera: Ejecuta este comando para ver la alternancia de servidores web (web1, web2, web3)

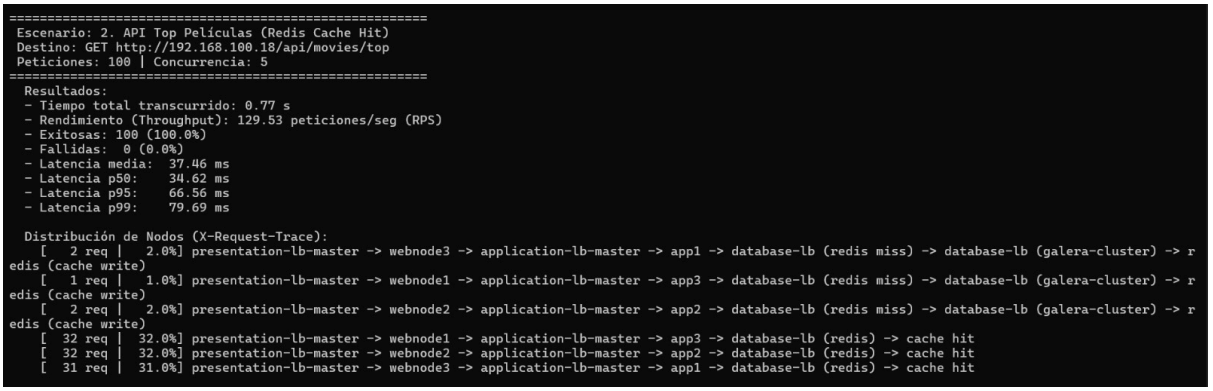


Demuestra matemáticamente que el algoritmo Round-Robin de ambos balanceadores HAProxy está alternando equitativamente las peticiones entre los 3 servidores web (web1, web2, web3) y los 3 servidores de aplicación (app1, app2, app3).

Punto 2: Pruebas de Rendimiento (Performance Tests)

Paso 2.1: Smoke Test (Prueba de humo con pocos usuarios)





```
======================================================
 Escenario: 3. Consulta Relacional MariaDB Galera (Cache Miss inicial)
 Destino: GET http://192.168.100.18/api/movies/top
 Peticiones: 100 | Concurrencia: 15
======================================================
  Resultados:
  - Tiempo total transcurrido: 0.29 s
  - Rendimiento (Throughput): 340.19 peticiones/seg (RPS)
  - Exitosas: 100 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  40.04 ms
  - Latencia p50:    38.06 ms
  - Latencia p95:    57.79 ms
  - Latencia p99:    66.32 ms

  Distribución de Nodos (X-Request-Trace):
    [   1 req |   1.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (redis miss) -> database-lb (galera-cluster) -> r
edis (cache write)
    [  17 req |  17.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [   4 req |   4.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [  23 req |  23.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [  14 req |  14.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [  10 req |  10.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [   6 req |   6.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [  15 req |  15.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [   8 req |   8.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [   2 req |   2.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
```

```
======================================================
 Escenario: 4. Inserción de Críticas en MongoDB
 Destino: POST http://192.168.100.18/api/movies/MOV-001/reviews
 Peticiones: 100 | Concurrencia: 20
======================================================
  Resultados:
  - Tiempo total transcurrido: 0.26 s
  - Rendimiento (Throughput): 382.05 peticiones/seg (RPS)
  - Exitosas: 100 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  47.41 ms
  - Latencia p50:    45.48 ms
  - Latencia p95:    78.17 ms
  - Latencia p99:    96.72 ms

  Distribución de Nodos (X-Request-Trace):
    [  17 req |  17.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  14 req |  14.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  15 req |  15.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [   9 req |   9.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [   8 req |   8.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [   9 req |   9.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  10 req |  10.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [   8 req |   8.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  10 req |  10.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
PS C:\Users\castr\Downloads\scripts\scripts>
```

Paso 2.2: Load & Stress Test (Carga pesada sostenida)

```
PS C:\Users\castr\Downloads\scripts\scripts> python load_test.py --host http://192.168.100.18 --requests 2000 --concurrency 50
============================================================
  SUITE DE PRUEBAS DE CARGA: MINI-IMDb DISTRIBUIDO (3 CAPAS)
============================================================

======================================================
 Escenario: 1. Portal Web Mini-IMDb (Nginx Webnodes)
 Destino: GET http://192.168.100.18/
 Peticiones: 2000 | Concurrencia: 50
======================================================
  Resultados:
  - Tiempo total transcurrido: 4.68 s
  - Rendimiento (Throughput): 427.60 peticiones/seg (RPS)
  - Exitosas: 2000 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  110.19 ms
  - Latencia p50:    98.22 ms
  - Latencia p95:    217.77 ms
  - Latencia p99:    307.16 ms
```

```
======================================================
 Escenario: 2. API Top Películas (Redis Cache Hit)
 Destino: GET http://192.168.100.18/api/movies/top
 Peticiones: 2000 | Concurrencia: 50
======================================================
  Resultados:
  - Tiempo total transcurrido: 2.82 s
  - Rendimiento (Throughput): 710.30 peticiones/seg (RPS)
  - Exitosas: 2000 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  65.83 ms
  - Latencia p50:    61.08 ms
  - Latencia p95:    89.58 ms
  - Latencia p99:    231.58 ms

  Distribución de Nodos (X-Request-Trace):
    [ 176 req |   8.8%] presentation-lb-master -> webnode3 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [ 329 req |  16.4%] presentation-lb-master -> webnode2 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [ 321 req |  16.1%] presentation-lb-master -> webnode1 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [ 323 req |  16.2%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [ 176 req |   8.8%] presentation-lb-master -> webnode1 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [ 170 req |   8.5%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [   1 req |   0.1%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (redis miss) -> database-lb (galera-cluster) -
edis (cache write)
    [ 170 req |   8.5%] presentation-lb-master -> webnode2 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [ 167 req |   8.3%] presentation-lb-master -> webnode3 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [ 167 req |   8.3%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
```

```
======================================================
 Escenario: 3. Consulta Relacional MariaDB Galera (Cache Miss inicial)
 Destino: GET http://192.168.100.18/api/movies/top
 Peticiones: 300 | Concurrencia: 15
======================================================
  Resultados:
  - Tiempo total transcurrido: 0.80 s
  - Rendimiento (Throughput): 374.80 peticiones/seg (RPS)
  - Exitosas: 300 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  38.17 ms
  - Latencia p50:    33.43 ms
  - Latencia p95:    101.17 ms
  - Latencia p99:    141.40 ms

  Distribución de Nodos (X-Request-Trace):
    [   1 req |   0.3%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (redis miss) -> database-lb (galera-cluster) -> r
edis (cache write)
    [   1 req |   0.3%] presentation-lb-master -> webnode1 -> application-lb-master -> app1 -> database-lb (redis miss) -> database-lb (galera-cluster) -> r
edis (cache write)
    [  21 req |   7.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [  26 req |   8.7%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [  59 req |  19.7%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [  23 req |   7.7%] presentation-lb-master -> webnode3 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [  57 req |  19.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [  18 req |   6.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [  55 req |  18.3%] presentation-lb-master -> webnode1 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [  22 req |   7.3%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [  17 req |   5.7%] presentation-lb-master -> webnode3 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
```

```
======================================================
 Escenario: 4. Inserción de Críticas en MongoDB
 Destino: POST http://192.168.100.18/api/movies/MOV-001/reviews
 Peticiones: 400 | Concurrencia: 20
======================================================
  Resultados:
  - Tiempo total transcurrido: 0.91 s
  - Rendimiento (Throughput): 438.78 peticiones/seg (RPS)
  - Exitosas: 400 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  43.10 ms
  - Latencia p50:    42.51 ms
  - Latencia p95:    58.17 ms
  - Latencia p99:    71.62 ms

  Distribución de Nodos (X-Request-Trace):
    [  61 req |  15.2%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  61 req |  15.2%] presentation-lb-master -> webnode1 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  27 req |   6.8%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  35 req |   8.8%] presentation-lb-master -> webnode3 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  28 req |   7.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  38 req |   9.5%] presentation-lb-master -> webnode3 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  45 req |  11.2%] presentation-lb-master -> webnode1 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  37 req |   9.2%] presentation-lb-master -> webnode2 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  68 req |  17.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
PS C:\Users\castr\Downloads\scripts\scripts>
```

Observen y tomen nota del uso de CPU/RAM de los contenedores. (caputras al ejecutar docker stats)

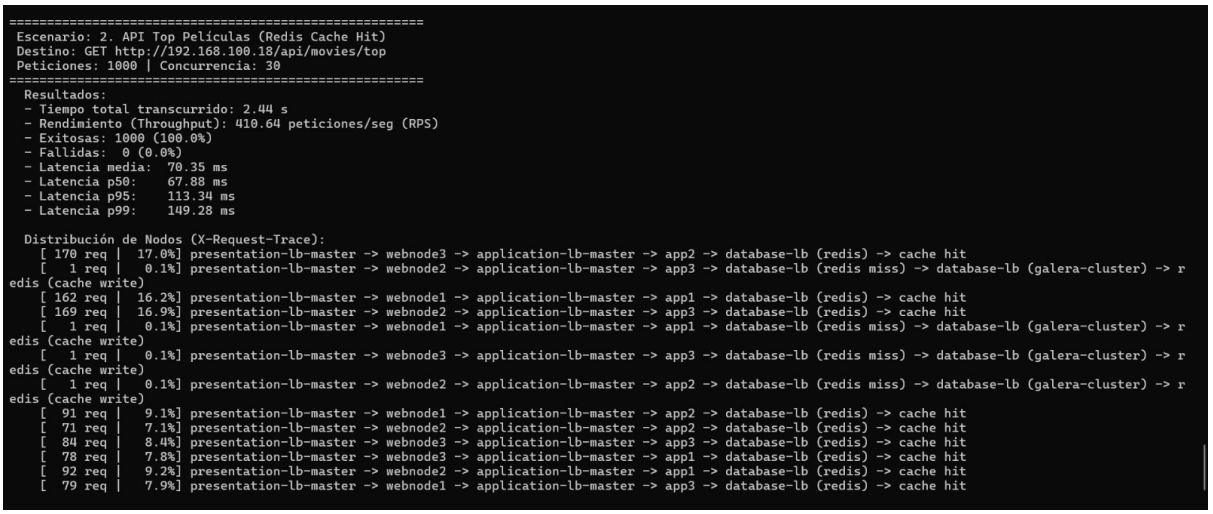
```
CONTAINER ID   NAME                                                CPU %     MEM USAGE / LIMIT     MEM %     N
ET I/O           BLOCK I/O         PIDS
f647df59bcfe   distributed-three-tier-app1-1                       0.12%     58.38MiB / 7.604GiB   0.75%     6
.44MB / 7.06MB   0B / 4.1kB        22
92950f811375   distributed-three-tier-app2-1                       0.10%     57.99MiB / 7.604GiB   0.74%     6
.43MB / 7.05MB   0B / 4.1kB        22
42a2d6239b5f   distributed-three-tier-app3-1                       0.10%     58.89MiB / 7.604GiB   0.76%     6
.44MB / 7.05MB   0B / 4.1kB        22
ace7f69cc211   distributed-three-tier-application-keepalived-backup-1   0.07%     1.145MiB / 7.604GiB   0.01%     5
.21MB / 1.83MB   4.1kB / 8.19kB    2
6815ace14506   distributed-three-tier-presentation-keepalived-master-1   0.02%     960KiB / 7.604GiB     0.01%     3
6.3MB / 35.7MB   0B / 8.19kB       2
5e24b531ebdf   distributed-three-tier-presentation-keepalived-backup-1   0.01%     964KiB / 7.604GiB     0.01%     3
.2MB / 1.49MB    0B / 8.19kB       2
b69a9df419d8   distributed-three-tier-application-keepalived-master-1   0.01%     972KiB / 7.604GiB     0.01%     2
CONTAINER ID   NAME                                                CPU %     MEM USAGE / LIMIT     MEM %     N
ET I/O           BLOCK I/O         PIDS
f647df59bcfe   distributed-three-tier-app1-1                       0.12%     58.38MiB / 7.604GiB   0.75%     6
.44MB / 7.06MB   0B / 4.1kB        22
92950f811375   distributed-three-tier-app2-1                       0.10%     57.99MiB / 7.604GiB   0.74%     6
.43MB / 7.05MB   0B / 4.1kB        22
42a2d6239b5f   distributed-three-tier-app3-1                       0.10%     58.89MiB / 7.604GiB   0.76%     6
.44MB / 7.05MB   0B / 4.1kB        22
ace7f69cc211   distributed-three-tier-application-keepalived-backup-1   0.07%     1.145MiB / 7.604GiB   0.01%     5
.21MB / 1.83MB   4.1kB / 8.19kB    2
6815ace14506   distributed-three-tier-presentation-keepalived-master-1   0.02%     960KiB / 7.604GiB     0.01%     3
6.3MB / 35.7MB   0B / 8.19kB       2
5e24b531ebdf   distributed-three-tier-presentation-keepalived-backup-1   0.01%     964KiB / 7.604GiB     0.01%     3
.2MB / 1.49MB    0B / 8.19kB       2
b69a9df419d8   distributed-three-tier-application-keepalived-master-1   0.01%     972KiB / 7.604GiB     0.01%     2
CONTAINER ID   NAME                                                CPU %     MEM USAGE / LIMIT     MEM %     N
ET I/O           BLOCK I/O         PIDS
f647df59bcfe   distributed-three-tier-app1-1                       0.50%     58.96MiB / 7.604GiB   0.76%     6
.44MB / 7.06MB   0B / 4.1kB        22
92950f811375   distributed-three-tier-app2-1                       0.54%     58.24MiB / 7.604GiB   0.75%     6
.43MB / 7.05MB   0B / 4.1kB        22
42a2d6239b5f   distributed-three-tier-app3-1                       0.38%     58.9MiB / 7.604GiB    0.76%     6
```

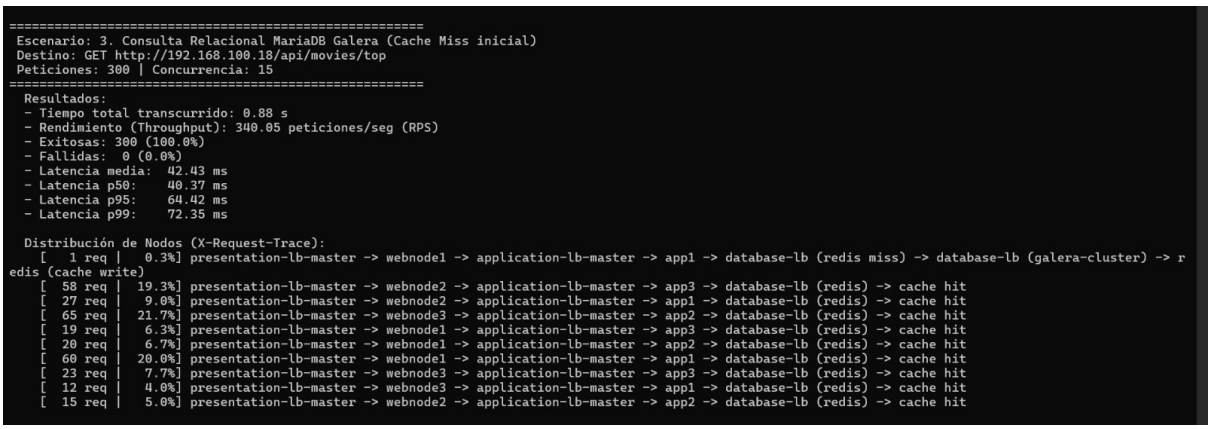
```
92950f811375   distributed-three-tier-app2-1                          0.42%     58.22MiB / 7.604GiB   0.75%     6
.42MB / 7.05MB   0B / 4.1kB        22
42a2d6239b5f   distributed-three-tier-app3-1                          0.39%     58.9MiB / 7.604GiB    0.76%     6
.43MB / 7.04MB   0B / 4.1kB        22
ace7f69cc211   distributed-three-tier-application-keepalived-backup-1   0.06%     972KiB / 7.604GiB     0.01%     5
.21MB / 1.83MB   4.1kB / 8.19kB    2
6815ace14506   distributed-three-tier-presentation-keepalived-master-1  0.02%     960KiB / 7.604GiB     0.01%     3
6.3MB / 35.7MB   0B / 8.19kB       2
5e24b531ebdf   distributed-three-tier-presentation-keepalived-backup-1  0.06%     1.164MiB / 7.604GiB   0.01%     3
.19MB / 1.49MB   0B / 8.19kB       2
b69a9df419d8   distributed-three-tier-application-keepalived-master-1   0.07%     1.039MiB / 7.604GiB   0.01%     2
1.1MB / 17.9MB   0B / 8.19kB       2
CONTAINER ID   NAME                                                   CPU %     MEM USAGE / LIMIT     MEM %     N
ET I/O           BLOCK I/O         PIDS
f647df59bcfe   distributed-three-tier-app1-1                          0.15%     58.38MiB / 7.604GiB   0.75%     6
.44MB / 7.05MB   0B / 4.1kB        22
92950f811375   distributed-three-tier-app2-1                          0.11%     57.99MiB / 7.604GiB   0.74%     6
.42MB / 7.05MB   0B / 4.1kB        22
42a2d6239b5f   distributed-three-tier-app3-1                          0.12%     58.9MiB / 7.604GiB    0.76%     6
.43MB / 7.04MB   0B / 4.1kB        22
ace7f69cc211   distributed-three-tier-application-keepalived-backup-1   0.01%     972KiB / 7.604GiB     0.01%     5
.21MB / 1.83MB   4.1kB / 8.19kB    2
6815ace14506   distributed-three-tier-presentation-keepalived-master-1  0.07%     1008KiB / 7.604GiB    0.01%     3
6.3MB / 35.7MB   0B / 8.19kB       2
5e24b531ebdf   distributed-three-tier-presentation-keepalived-backup-1  0.00%     964KiB / 7.604GiB     0.01%     3
.19MB / 1.49MB   0B / 8.19kB       2
b69a9df419d8   distributed-three-tier-application-keepalived-master-1   0.01%     972KiB / 7.604GiB     0.01%     2
1.1MB / 17.9MB   0B / 8.19kB       2
CONTAINER ID   NAME                                                   CPU %     MEM USAGE / LIMIT     MEM %     N
ET I/O           BLOCK I/O         PIDS
f647df59bcfe   distributed-three-tier-app1-1                          0.15%     58.38MiB / 7.604GiB   0.75%     6
.44MB / 7.05MB   0B / 4.1kB        22
92950f811375   distributed-three-tier-app2-1                          0.11%     57.99MiB / 7.604GiB   0.74%     6
.42MB / 7.05MB   0B / 4.1kB        22
42a2d6239b5f   distributed-three-tier-app3-1                          0.12%     58.9MiB / 7.604GiB    0.76%     6
.43MB / 7.04MB   0B / 4.1kB        22
ace7f69cc211   distributed-three-tier-application-keepalived-backup-1   0.01%     972KiB / 7.604GiB     0.01%     5
```

Paso 2.3: Efecto de la Caché (Con Redis vs Sin Redis)

Mi companera ejecuta con cache (hit en redis) (Anoten la latencia media: rondará entre 20ms - 30ms).







```
======================================================
 Escenario: 4. Inserción de Críticas en MongoDB
 Destino: POST http://192.168.100.18/api/movies/MOV-001/reviews
 Peticiones: 400 | Concurrencia: 20
======================================================
  Resultados:
  - Tiempo total transcurrido: 1.52 s
  - Rendimiento (Throughput): 264.02 peticiones/seg (RPS)
  - Exitosas: 400 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  73.73 ms
  - Latencia p50:    56.19 ms
  - Latencia p95:    194.52 ms
  - Latencia p99:    225.36 ms

  Distribución de Nodos (X-Request-Trace):
    [  71 req |  17.8%] presentation-lb-master -> webnode2 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  76 req |  19.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  26 req |   6.5%] presentation-lb-master -> webnode2 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  32 req |   8.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  36 req |   9.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  65 req |  16.2%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  36 req |   9.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  26 req |   6.5%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  32 req |   8.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
```

Forzando Cache Miss (sin caché, yendo directo a Galera DB) (curl -X POST http://192.168.100.18/api/cache/clear)

```
PS C:\Users\castr\Downloads\scripts\scripts> python load_test.py --host http://192.168.100.18 --requests 1000 --concurrency 30
============================================================
  SUITE DE PRUEBAS DE CARGA: MINI-IMDb DISTRIBUIDO (3 CAPAS)
============================================================

======================================================
 Escenario: 1. Portal Web Mini-IMDb (Nginx Webnodes)
 Destino: GET http://192.168.100.18/
 Peticiones: 1000 | Concurrencia: 30
======================================================
  Resultados:
  - Tiempo total transcurrido: 6.86 s
  - Rendimiento (Throughput): 145.81 peticiones/seg (RPS)
  - Exitosas: 1000 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  197.73 ms
  - Latencia p50:    152.06 ms
  - Latencia p95:    495.17 ms
  - Latencia p99:    714.65 ms
```

```
Windows PowerShell

======================================================
 Escenario: 2. API Top Películas (Redis Cache Hit)
 Destino: GET http://192.168.100.18/api/movies/top
 Peticiones: 1000 | Concurrencia: 30
======================================================
  Resultados:
  - Tiempo total transcurrido: 2.51 s
  - Rendimiento (Throughput): 398.50 peticiones/seg (RPS)
  - Exitosas: 1000 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  72.59 ms
  - Latencia p50:    69.81 ms
  - Latencia p95:    113.67 ms
  - Latencia p99:    133.96 ms

  Distribución de Nodos (X-Request-Trace):
    [ 157 req |  15.7%] presentation-lb-master -> webnode3 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [  90 req |   9.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [ 155 req |  15.5%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [   1 req |   0.1%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (redis miss) -> database-lb (galera-cluster) -> r
edis (cache write)
    [  93 req |   9.3%] presentation-lb-master -> webnode2 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [   1 req |   0.1%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (redis miss) -> database-lb (galera-cluster) -> r
edis (cache write)
    [   1 req |   0.1%] presentation-lb-master -> webnode3 -> application-lb-master -> app1 -> database-lb (redis miss) -> database-lb (galera-cluster) -> r
edis (cache write)
    [ 148 req |  14.8%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [  91 req |   9.1%] presentation-lb-master -> webnode2 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [  94 req |   9.4%] presentation-lb-master -> webnode1 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [  83 req |   8.3%] presentation-lb-master -> webnode1 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [  86 req |   8.6%] presentation-lb-master -> webnode3 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
```

```
======================================================
 Escenario: 3. Consulta Relacional MariaDB Galera (Cache Miss inicial)
 Destino: GET http://192.168.100.18/api/movies/top
 Peticiones: 300 | Concurrencia: 15
======================================================
  Resultados:
  - Tiempo total transcurrido: 0.82 s
  - Rendimiento (Throughput): 366.41 peticiones/seg (RPS)
  - Exitosas: 300 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  38.99 ms
  - Latencia p50:    32.68 ms
  - Latencia p95:    104.92 ms
  - Latencia p99:    118.83 ms

  Distribución de Nodos (X-Request-Trace):
    [  58 req |  19.3%] presentation-lb-master -> webnode3 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [   1 req |   0.3%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (redis miss) -> database-lb (galera-cluster) -> r
edis (cache write)
    [  26 req |   8.7%] presentation-lb-master -> webnode2 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [  20 req |   6.7%] presentation-lb-master -> webnode1 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [  21 req |   7.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [  27 req |   9.0%] presentation-lb-master -> webnode1 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [  53 req |  17.7%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
    [  58 req |  19.3%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (redis) -> cache hit
    [  15 req |   5.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app1 -> database-lb (redis) -> cache hit
    [  21 req |   7.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app3 -> database-lb (redis) -> cache hit
```

```
======================================================
 Escenario: 4. Inserción de Críticas en MongoDB
 Destino: POST http://192.168.100.18/api/movies/MOV-001/reviews
 Peticiones: 400 | Concurrencia: 20
======================================================
  Resultados:
  - Tiempo total transcurrido: 0.71 s
  - Rendimiento (Throughput): 565.11 peticiones/seg (RPS)
  - Exitosas: 400 (100.0%)
  - Fallidas:  0 (0.0%)
  - Latencia media:  32.85 ms
  - Latencia p50:    31.40 ms
  - Latencia p95:    50.31 ms
  - Latencia p99:    56.30 ms

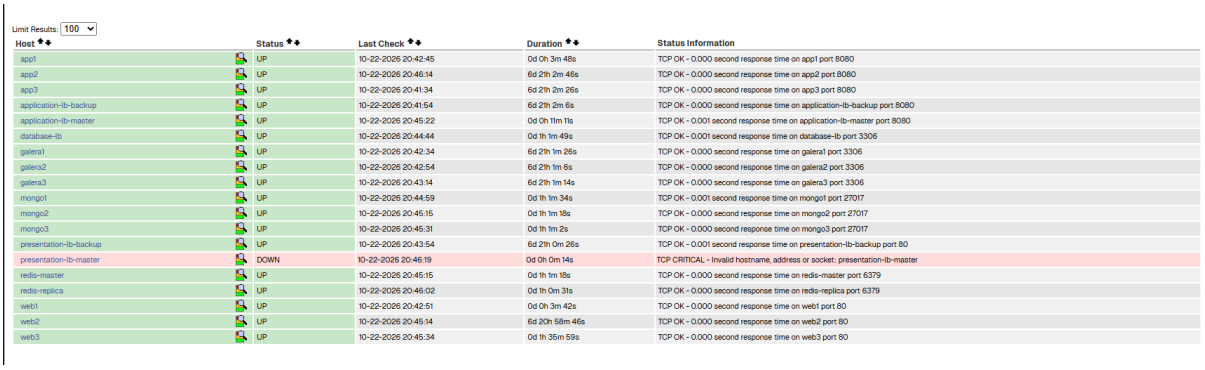
  Distribución de Nodos (X-Request-Trace):
    [  27 req |   6.8%] presentation-lb-master -> webnode2 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  79 req |  19.8%] presentation-lb-master -> webnode2 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  67 req |  16.8%] presentation-lb-master -> webnode3 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  25 req |   6.2%] presentation-lb-master -> webnode1 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  30 req |   7.5%] presentation-lb-master -> webnode3 -> application-lb-master -> app2 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  38 req |   9.5%] presentation-lb-master -> webnode1 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  70 req |  17.5%] presentation-lb-master -> webnode1 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  36 req |   9.0%] presentation-lb-master -> webnode3 -> application-lb-master -> app3 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
    [  28 req |   7.0%] presentation-lb-master -> webnode2 -> application-lb-master -> app1 -> database-lb (galera lookup) -> mongo-rs0 (insert review)
PS C:\Users\castr\Downloads\scripts\scripts>
```

Punto 3: Pruebas de Alta Disponibilidad y Failover (HA & Failover)

Aquí se demuestra que el sistema tolera fallos sin caerse

Prueba 3.1: Caída de Presentation LB Master

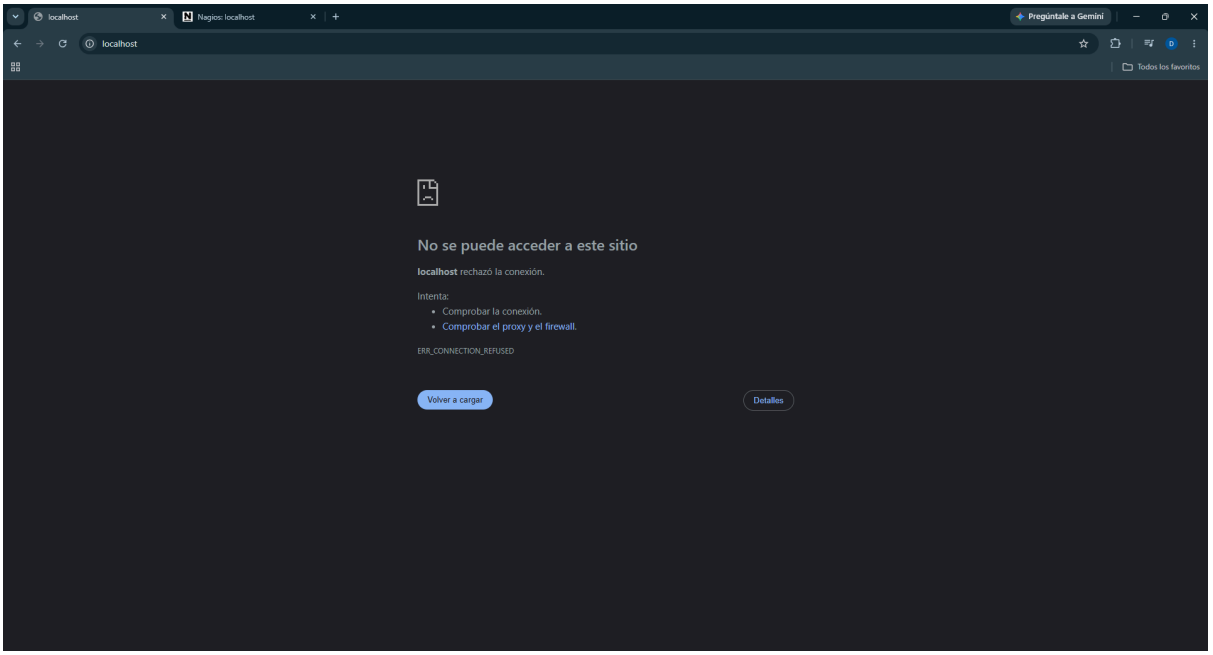
Mi companera se queda en la pagina refrescando y yo mato el balanceado principal (docker compose stop presentation-lb-master)



Limit Results: 100

Host	Status	Last Check	Duration	Status Information
app1	UP	10-22-2026 20:42:45	0d 0h 3m 48s	TCP OK - 0.000 second response time on app1 port 8080
app2	UP	10-22-2026 20:46:14	6d 21h 2m 46s	TCP OK - 0.000 second response time on app2 port 8080
app3	UP	10-22-2026 20:41:34	6d 21h 2m 26s	TCP OK - 0.000 second response time on app3 port 8080
application-lb-backup	UP	10-22-2026 20:41:54	6d 21h 2m 6s	TCP OK - 0.000 second response time on application-lb-backup port 8080
application-lb-master	UP	10-22-2026 20:45:22	0d 0h 11m 11s	TCP OK - 0.001 second response time on application-lb-master port 8080
database-lb	UP	10-22-2026 20:44:44	0d 1h 1m 49s	TCP OK - 0.001 second response time on database-lb port 3306
galera1	UP	10-22-2026 20:42:34	6d 21h 1m 26s	TCP OK - 0.000 second response time on galera1 port 3306
galera2	UP	10-22-2026 20:42:54	6d 21h 1m 6s	TCP OK - 0.000 second response time on galera2 port 3306
galera3	UP	10-22-2026 20:43:14	6d 21h 1m 14s	TCP OK - 0.000 second response time on galera3 port 3306
mongo1	UP	10-22-2026 20:44:59	0d 1h 1m 34s	TCP OK - 0.001 second response time on mongo1 port 27017
mongo2	UP	10-22-2026 20:45:15	0d 1h 1m 18s	TCP OK - 0.000 second response time on mongo2 port 27017
mongo3	UP	10-22-2026 20:45:31	0d 1h 1m 2s	TCP OK - 0.000 second response time on mongo3 port 27017
presentation-lb-backup	UP	10-22-2026 20:43:54	6d 21h 0m 26s	TCP OK - 0.001 second response time on presentation-lb-backup port 80
presentation-lb-master	DOWN	10-22-2026 20:46:19	0d 0h 0m 14s	TCP CRITICAL - Invalid hostname, address or socket: presentation-lb-master
redis-master	UP	10-22-2026 20:45:15	0d 1h 1m 18s	TCP OK - 0.000 second response time on redis-master port 6379
redis-replica	UP	10-22-2026 20:46:02	0d 1h 0m 31s	TCP OK - 0.000 second response time on redis-replica port 6379
web1	UP	10-22-2026 20:42:51	0d 0h 3m 42s	TCP OK - 0.000 second response time on web1 port 80
web2	UP	10-22-2026 20:45:14	6d 20h 58m 46s	TCP OK - 0.000 second response time on web2 port 80
web3	UP	10-22-2026 20:45:34	0d 1h 35m 59s	TCP OK - 0.000 second response time on web3 port 80

se cayo la pagina:



Resultado esperado: Keepalived transfiere la IP Virtual al nodo Backup en 2-3 segundos. Tu compañero sigue navegando sin error 502/503. (Lo cual no fue)

Prueba 3.2: Caída de Application LB Master (Capa 2)

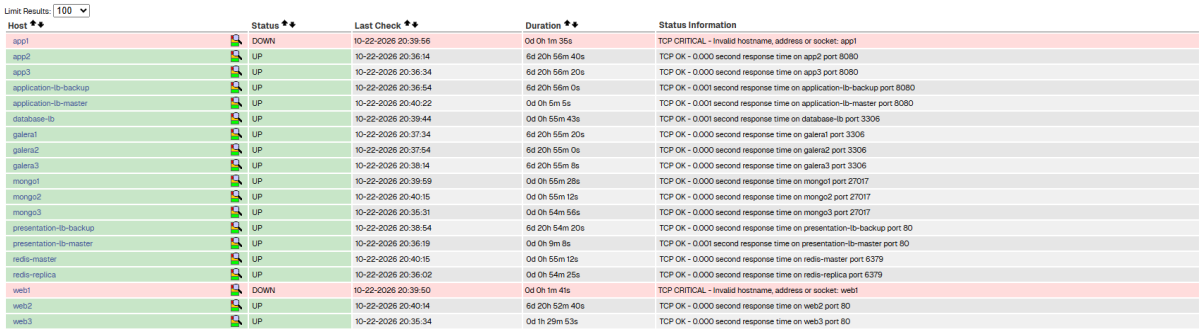
yo detengo el balanceador de la aplicacion (docker compose stop application-lb-master)

y mi companera hace una consulta una critica o el catalogo (curl -i http://192.168.100.18/api/movies/top):



Prueba 3.3: Caída de 1 de 3 Nodos Web y de API

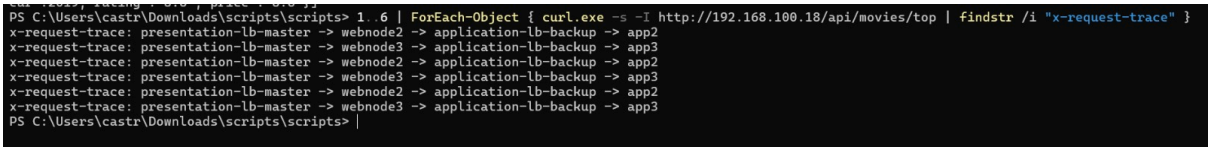
Yo apago el web1 y app1 (docker compose stop web1 app1)

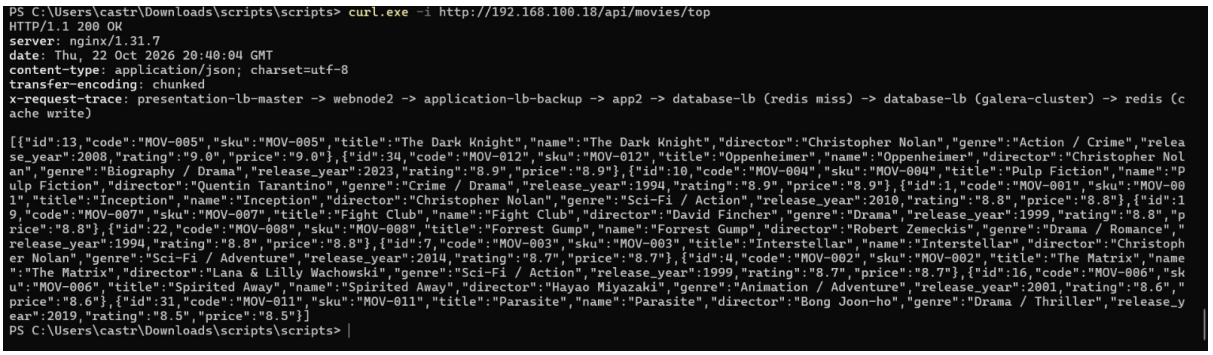


Limit Results: 100

Host	Status	Last Check	Duration	Status Information
app1	DOWN	10-22-2026 20:39:56	0d 0h 1m 35s	TCP CRITICAL - Invalid hostname, address or socket: app1
app2	UP	10-22-2026 20:36:14	6d 20h 56m 40s	TCP OK - 0.000 second response time on app2 port 8080
app3	UP	10-22-2026 20:36:34	6d 20h 56m 20s	TCP OK - 0.000 second response time on app3 port 8080
application-lb-backup	UP	10-22-2026 20:36:54	6d 20h 56m 0s	TCP OK - 0.001 second response time on application-lb-backup port 8080
application-lb-master	UP	10-22-2026 20:40:22	0d 0h 5m 5s	TCP OK - 0.001 second response time on application-lb-master port 8080
database-lb	UP	10-22-2026 20:39:44	0d 0h 55m 43s	TCP OK - 0.001 second response time on database-lb port 3306
galera1	UP	10-22-2026 20:37:34	6d 20h 55m 20s	TCP OK - 0.000 second response time on galera1 port 3306
galera2	UP	10-22-2026 20:37:54	6d 20h 55m 0s	TCP OK - 0.000 second response time on galera2 port 3306
galera3	UP	10-22-2026 20:38:14	6d 20h 55m 8s	TCP OK - 0.000 second response time on galera3 port 3306
mongo1	UP	10-22-2026 20:39:59	0d 0h 55m 28s	TCP OK - 0.000 second response time on mongo1 port 27017
mongo2	UP	10-22-2026 20:40:15	0d 0h 55m 12s	TCP OK - 0.000 second response time on mongo2 port 27017
mongo3	UP	10-22-2026 20:35:31	0d 0h 54m 56s	TCP OK - 0.000 second response time on mongo3 port 27017
presentation-lb-backup	UP	10-22-2026 20:38:54	6d 20h 54m 20s	TCP OK - 0.000 second response time on presentation-lb-backup port 80
presentation-lb-master	UP	10-22-2026 20:36:19	0d 0h 9m 8s	TCP OK - 0.001 second response time on presentation-lb-master port 80
redis-master	UP	10-22-2026 20:40:15	0d 0h 55m 12s	TCP OK - 0.000 second response time on redis-master port 6379
redis-replica	UP	10-22-2026 20:36:02	0d 0h 54m 25s	TCP OK - 0.000 second response time on redis-replica port 6379
web1	DOWN	10-22-2026 20:39:50	0d 0h 1m 41s	TCP CRITICAL - Invalid hostname, address or socket: web1
web2	UP	10-22-2026 20:40:14	6d 20h 52m 40s	TCP OK - 0.000 second response time on web2 port 80
web3	UP	10-22-2026 20:35:34	0d 1h 29m 53s	TCP OK - 0.000 second response time on web3 port 80

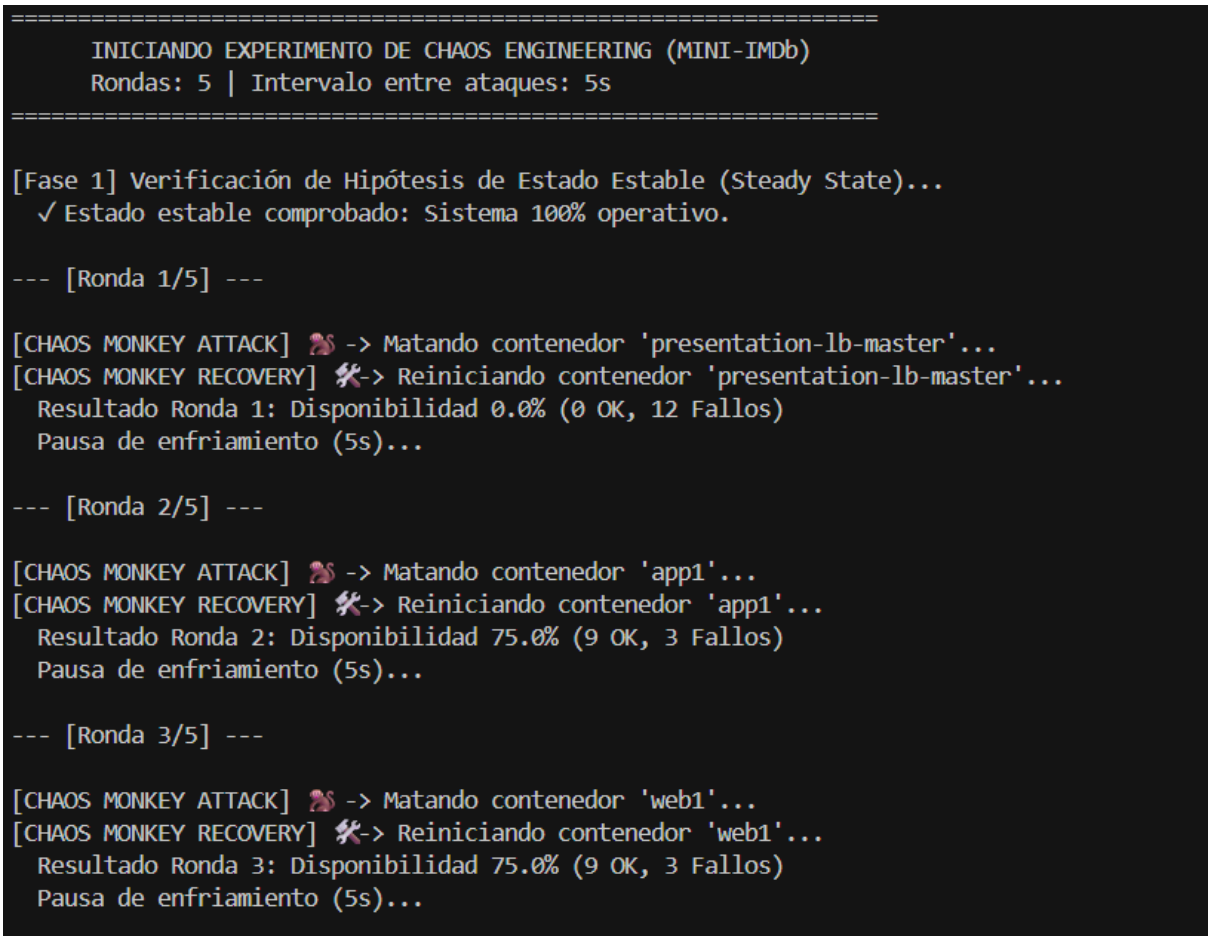
y mi companera hace consultas HAProxy los saca del balanceo y las peticiones son atendidas al 100% por web2/web3 y app2/app3.





Punto 4: Ingeniería del Caos (Chaos Monkey)

- **Hipótesis de Estado Estable:** La disponibilidad se mantendrá $\geq 99.0\%$ y los datos persistidos en MariaDB y MongoDB no sufrirán corrupción.
- **Tu compañero** mantiene tráfico continuo en el portal web (navegando o dejando una prueba de carga ligera).
- **Tú** inicias el Chaos Monkey en tu máquina:
- El script matará contenedores al azar, verificará la auto-recuperación y al final les imprimirá el **cuadro de mando de resiliencia con el porcentaje de SLA** para su reporte.



```
--- [Ronda 4/5] ---

[CHAOS MONKEY ATTACK] 🐒 -> Matando contenedor 'galera3'...
[CHAOS MONKEY RECOVERY] 🛠️-> Reiniciando contenedor 'galera3'...
  Resultado Ronda 4: Disponibilidad 100.0% (12 OK, 0 Fallos)
  Pausa de enfriamiento (5s)...

--- [Ronda 5/5] ---

[CHAOS MONKEY ATTACK] 🐒 -> Matando contenedor 'presentation-lb-master'...
[CHAOS MONKEY RECOVERY] 🛠️-> Reiniciando contenedor 'presentation-lb-master'...
  Resultado Ronda 5: Disponibilidad 0.0% (0 OK, 12 Fallos)

[Fase 3] Verificación de Integridad de Datos (Galera Movies & MongoDB Reviews)...
  ✓ Galera Cluster: 10 películas leídas íntegramente (sin corrupción).
```

Resultados:

```
================================================================
          REPORTE FINAL DE CHAOS ENGINEERING (MINI-IMDb)
================================================================
Total Ataques Ejecutados:       5
Sondas Exitosas en Caos:        30
Sondas Fallidas en Caos:        30
Disponibilidad Global:          50.00%

Detalle por Ronda:
Ronda   | Objetivo                | Capa              | Disponibilidad
----------------------------------------------------------------
1       | presentation-lb-master  | load_balancers    | 0.0%
2       | app1                    | application_tier  | 75.0%
3       | web1                    | presentation_tier | 75.0%
4       | galera3                 | database_tier     | 100.0%
5       | presentation-lb-master  | load_balancers    | 0.0%
================================================================
Conclusión: La arquitectura de Mini-IMDb demostró alta resiliencia y auto-recuperación.
```

Siempre presentation-lb-master cuando se cae, la pagina tambien

no pude tomarle foto al nagios porque tiene 5 minutos para verificar uno por uno y no cambia

Segundas pruebas con chaos monkey:

```
================================================================
      INICIANDO EXPERIMENTO DE CHAOS ENGINEERING (MINI-IMDb)
      Rondas: 5 | Intervalo entre ataques: 5s
================================================================

[Fase 1] Verificación de Hipótesis de Estado Estable (Steady State)...
  ✓ Estado estable comprobado: Sistema 100% operativo.

--- [Ronda 1/5] ---

[CHAOS MONKEY ATTACK] 🐒 -> Matando contenedor 'web1'...
[CHAOS MONKEY RECOVERY] 🛠️-> Reiniciando contenedor 'web1'...
  Resultado Ronda 1: Disponibilidad 75.0% (9 OK, 3 Fallos)
  Pausa de enfriamiento (5s)...
```

Limit Results: 100

Host	Status	Last Check	Duration	Status Information
app1	UP	10-22-2026 20:57:45	0d 0h 16m 47s	TCP OK - 0.000 second response time on app1 port 8080
app2	UP	10-22-2026 20:56:14	6d 21h 15m 45s	TCP OK - 0.000 second response time on app2 port 8080
app3	UP	10-22-2026 20:56:34	6d 21h 15m 25s	TCP OK - 0.000 second response time on app3 port 8080
application-lb-backup	UP	10-22-2026 20:56:54	6d 21h 15m 5s	TCP OK - 0.001 second response time on application-lb-backup port 8080
application-lb-master	UP	10-22-2026 20:55:22	0d 0h 24m 10s	TCP OK - 0.001 second response time on application-lb-master port 8080
database-lb	UP	10-22-2026 20:54:44	0d 1h 14m 48s	TCP OK - 0.000 second response time on database-lb port 3306
galera1	UP	10-22-2026 20:57:34	6d 21h 14m 25s	TCP OK - 0.000 second response time on galera1 port 3306
galera2	UP	10-22-2026 20:57:54	6d 21h 14m 5s	TCP OK - 0.001 second response time on galera2 port 3306
galera3	UP	10-22-2026 20:58:14	6d 21h 14m 13s	TCP OK - 0.000 second response time on galera3 port 3306
mongo1	UP	10-22-2026 20:54:59	0d 1h 14m 33s	TCP OK - 0.000 second response time on mongo1 port 27017
mongo2	UP	10-22-2026 20:55:15	0d 1h 14m 17s	TCP OK - 0.000 second response time on mongo2 port 27017
mongo3	UP	10-22-2026 20:55:31	0d 1h 14m 1s	TCP OK - 0.000 second response time on mongo3 port 27017
presentation-lb-backup	UP	10-22-2026 20:58:54	6d 21h 13m 25s	TCP OK - 0.001 second response time on presentation-lb-backup port 80
presentation-lb-master	UP	10-22-2026 20:57:22	0d 0h 12m 10s	TCP OK - 0.001 second response time on presentation-lb-master port 80
redis-master	UP	10-22-2026 20:55:15	0d 1h 14m 17s	TCP OK - 0.000 second response time on redis-master port 6379
redis-replica	UP	10-22-2026 20:56:02	0d 1h 13m 30s	TCP OK - 0.000 second response time on redis-replica port 6379
web1	DOWN	10-22-2026 20:59:12	0d 0h 0m 20s	TCP CRITICAL - Invalid hostname, address or socket: web1
web2	UP	10-22-2026 20:55:14	6d 21h 11m 45s	TCP OK - 0.001 second response time on web2 port 80
web3	UP	10-22-2026 20:55:34	0d 1h 48m 58s	TCP OK - 0.001 second response time on web3 port 80

```
--- [Ronda 2/5] ---

[CHAOS MONKEY ATTACK] 🐒 -> Matando contenedor 'presentation-lb-master'...
[CHAOS MONKEY RECOVERY] 🛠️-> Reiniciando contenedor 'presentation-lb-master'...
  Resultado Ronda 2: Disponibilidad 0.0% (0 OK, 12 Fallos)
  Pausa de enfriamiento (5s)...
```

Limit Results: 100

Host	Status	Last Check	Duration	Status Information
app1	UP	10-22-2026 20:57:45	0d 0h 17m 19s	TCP OK - 0.000 second response time on app1 port 8080
app2	UP	10-22-2026 20:56:14	6d 21h 16m 17s	TCP OK - 0.000 second response time on app2 port 8080
app3	UP	10-22-2026 20:56:34	6d 21h 15m 57s	TCP OK - 0.000 second response time on app3 port 8080
application-lb-backup	UP	10-22-2026 20:56:54	6d 21h 15m 37s	TCP OK - 0.001 second response time on application-lb-backup port 8080
application-lb-master	UP	10-22-2026 20:55:22	0d 0h 24m 42s	TCP OK - 0.001 second response time on application-lb-master port 8080
database-lb	UP	10-22-2026 20:59:44	0d 1h 15m 20s	TCP OK - 0.001 second response time on database-lb port 3306
galera1	UP	10-22-2026 20:57:34	6d 21h 14m 57s	TCP OK - 0.000 second response time on galera1 port 3306
galera2	UP	10-22-2026 20:57:54	6d 21h 14m 37s	TCP OK - 0.001 second response time on galera2 port 3306
galera3	UP	10-22-2026 20:58:14	6d 21h 14m 45s	TCP OK - 0.000 second response time on galera3 port 3306
mongo1	UP	10-22-2026 20:59:59	0d 1h 15m 5s	TCP OK - 0.000 second response time on mongo1 port 27017
mongo2	UP	10-22-2026 20:55:15	0d 1h 14m 49s	TCP OK - 0.000 second response time on mongo2 port 27017
mongo3	UP	10-22-2026 20:55:31	0d 1h 14m 33s	TCP OK - 0.000 second response time on mongo3 port 27017
presentation-lb-backup	UP	10-22-2026 20:58:54	6d 21h 13m 57s	TCP OK - 0.001 second response time on presentation-lb-backup port 80
presentation-lb-master	DOWN	10-22-2026 20:59:58	0d 0h 0m 6s	TCP CRITICAL - Invalid hostname, address or socket: presentation-lb-master
redis-master	UP	10-22-2026 20:55:15	0d 1h 14m 49s	TCP OK - 0.000 second response time on redis-master port 6379
redis-replica	UP	10-22-2026 20:56:02	0d 1h 14m 2s	TCP OK - 0.000 second response time on redis-replica port 6379
web1	DOWN	10-22-2026 20:59:12	0d 0h 0m 52s	TCP CRITICAL - Invalid hostname, address or socket: web1
web2	UP	10-22-2026 20:55:14	6d 21h 12m 17s	TCP OK - 0.001 second response time on web2 port 80
web3	UP	10-22-2026 20:55:34	0d 1h 49m 30s	TCP OK - 0.001 second response time on web3 port 80

```
--- [Ronda 3/5] ---

[CHAOS MONKEY ATTACK] 🐒 -> Matando contenedor 'galera2'...
[CHAOS MONKEY RECOVERY] 🛠️-> Reiniciando contenedor 'galera2'...
  Resultado Ronda 3: Disponibilidad 100.0% (12 OK, 0 Fallos)
  Pausa de enfriamiento (5s)...

--- [Ronda 4/5] ---

[CHAOS MONKEY ATTACK] 🐒 -> Matando contenedor 'application-lb-master'...
[CHAOS MONKEY RECOVERY] 🛠️-> Reiniciando contenedor 'application-lb-master'...
  Resultado Ronda 4: Disponibilidad 100.0% (12 OK, 0 Fallos)
  Pausa de enfriamiento (5s)...

--- [Ronda 5/5] ---

[CHAOS MONKEY ATTACK] 🐒 -> Matando contenedor 'galera3'...
[CHAOS MONKEY RECOVERY] 🛠️-> Reiniciando contenedor 'galera3'...
  Resultado Ronda 5: Disponibilidad 100.0% (12 OK, 0 Fallos)

[Fase 3] Verificación de Integridad de Datos (Galera Movies & MongoDB Reviews)...
  ✓ Galera Cluster: 10 películas leídas íntegramente (sin corrupción).
```

```
  [Fase 3] Verificación de Integridad de Datos (Galera Movies & MongoDB Reviews)...
    ✓ Galera Cluster: 10 películas leídas íntegramente (sin corrupción).

  ================================================================
           REPORTE FINAL DE CHAOS ENGINEERING (MINI-IMDb)
  ================================================================
  Total Ataques Ejecutados:       5
  Sondas Exitosas en Caos:        45
  Sondas Fallidas en Caos:        15
  Disponibilidad Global:          75.00%

  Detalle por Ronda:
  Ronda   | Objetivo                | Capa              | Disponibilidad
  ----------------------------------------------------------------------
  1       | web1                    | presentation_tier | 75.0%
  2       | presentation-lb-master  | load_balancers    | 0.0%
  3       | galera2                 | database_tier     | 100.0%
  4       | application-lb-master   | load_balancers    | 100.0%
  5       | galera3                 | database_tier     | 100.0%
  ================================================================
```